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Analysis I – Übungsblatt 9

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Aufgabe 1 (6 P.)
Seien $(X, \|\cdot\|)$ ein normierter Raum und $M \subset X$. Beweisen Sie, dass folgende Aussagen äquivalent sind (Satz 4.2.14 der Vorlesung):

(a) M ist abgeschlossen.

(b) M enthält alle seine Häufungspunkte.

(c) Für jede konvergente Folge $(x_n)_{n\in\mathbb{N}}$ mit $x_n \in M$ falls alle $n \in \mathbb{N}$ ist $\lim_{n\to\infty} x_n \in M$.

Lösung. "(a)⇒(b)": Let M be closed, then $X \setminus M$ is open. Let $x \in X \setminus M$, we want to show that x is not an accumulation point of M. Since $X \setminus M$ is open, there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subset X \setminus M$. This means that $B(x, \varepsilon) \cap M = \emptyset$. Therefore x cannot be an accumulation point.

"(b)⇒(c)": Let M be such that it contains all its accumulation points and let $(x_n)_{n\in\mathbb{N}} \subset M$ be a convergent sequence, with limit $x \in X$. We need to show that $x \in M$. We differ two cases:

(i) $\exists \tilde{n} \in \mathbb{N}$ such that $x_{\tilde{n}} = x$: Since $x_{\tilde{n}} \in M$ we conclude $x \in M$.

(ii) $x_n \neq x \ \forall n \in \mathbb{N}$: By Theorem 4.2.13 this is equivalent to say that x is an accumulation point. Since M contains all its accumulation points, we have $x \in M$.

"(c) ⇒ (a)": Let x be a BP (Berührunspunkt) of M. By Theorem 4.2.13, there exists a sequence $(x_n)_{n\in\mathbb{N}} \subset M$ such that $\lim_{n\to\infty} x_n = x$. By assumption (c), we have $x \in M$. Since x was an arbitrary BP, this means $\overline{M} := \{x \in X : x \text{ is a BP of } M\} \subset M$. Moreover $M \subset \overline{M}$ by Theorem 4.2.12 (1). Thus

$$M = \overline{M}.$$

By Theorem 4.2.12 (2), M is closed. □

Aufgabe 2 (3+3+3 P.)
Seien $(X, \|\cdot\|)$ ein normierter Raum und $\emptyset \neq M \subset X$. Sind die folgenden Aussagen richtig oder falsch? Beweisen oder widerlegen Sie sie (durch Gegenbeispiele).

(a) $\overset{\circ}{\overline{M}} = \overset{\circ}{M}$

(b) $\overline{\overset{\circ}{M}} = \overline{M}$

(c) $\overset{\circ}{\overline{M}} = \overline{\overset{\circ}{M}}$

Lösung.

(a) <u>Claim:</u> $\overset{\circ}{\overline{M}} \neq \overset{\circ}{M}$

We can prove this by considering the counterexample $M := B(0,1) \setminus \{0\} = \{x \in X : 0 < \|x\| < 1\}$ (example 4.2.22 (2) of the lecture notes). Recall that by definition $\overset{\circ}{M} = \{x \in M : \exists U \in \mathcal{U}(x), U \subset M\}$ and hence, $\overset{\circ}{M} \subset M$.

We now show the other inclusion: Let $x \in M$. Then $\exists r > 0$ such that $0 < r < \|x\| < 1 - r < 1$. Hence, $B(x,r) \subset M$ and $x \in \overset{\circ}{M}$. This means every point in M is also in $\overset{\circ}{M}$, thus $M = \overset{\circ}{M}$.

We claim that $\overline{M} = \{x \in X : \|x\| \leq 1\}$. Let $(x_n)_{n\in\mathbb{N}} \subset M$ be a convergent sequence and call its limit $x \in X$. As $0 < \|x_n\| < 1 \ \forall n \in \mathbb{N}$, it follows that $0 \leq \|x\| \leq 1$. Hence, $\overline{M} \subset \{x \in X : \|x\| \leq 1\}$. Assume (by contradiction) there is a $y \in \{x \in X : \|x\| \leq 1\} \setminus \overline{M}$. As $M \subset \overline{M}$, either $y = 0$ or $\|y\| = 1$.

To exclude the case $y = 0$, we construct a sequence in M converging to 0. Namely, let $z \in M$ be arbitrary and consider $z_n := \frac{1}{n} z$. As $0 < \|z\| < 1$, also $0 < \|z_n\| = \frac{1}{n}\|z\| < 1$ and thus, $z_n \in M$ for all $n \geq 1$. Moreover, $\|z_n\| = \frac{1}{n}\|z\| \to$ as $n \to \infty$, hence $z_n \to 0$.

In the same way we exclude the case $\|y\| = 1$: Consider $a_n := \frac{n}{n+1} y$. Then $0 < \|a_n\| = \frac{n}{n+1}\|y\| = \frac{n}{n+1} < 1$ for all $n \geq 1$. Moreover

$$\|a_n - y\| = \left\| \frac{n}{n+1} y - y \right\| = \frac{1}{n+1}\|y\| = \frac{1}{n+1} \overset{n\to\infty}{\longrightarrow} 0.$$

Hence, $a_n \overset{n\to\infty}{\longrightarrow} y$.

Now we claim $\overset{\circ}{\overline{M}} = \{x \in X : \|x\| < 1\}$. By the same reasoning as before, the latter set is open. Now we show it is the biggest open set contained in $\overline{M}$. Assume (by contradiction) there is a bigger one, thus there is $x \in \overset{\circ}{\overline{M}}$ with $\|x\| \geq 1$. Let $r > 0$ be arbitrary, then $B(x,r)$ contains points with norm greater than 1, thus not lying in $\overline{M}$. Therefore $x \notin \overset{\circ}{\overline{M}}$. So in total we have

$$\overset{\circ}{\overline{M}} = \{x \in X : \|x\| < 1\} \neq M = \overset{\circ}{M}.$$

(b) <u>Claim:</u> $\overline{\overset{\circ}{M}} \neq \overline{M}$

We can prove this by considering the counterexample $M = \{x\}$ for some $x \in X$.

As M is closed (Satz 4.2.12 (2)) it follows that $\overline{M} = M = \{x\}$. Moreover, for all $r > 0$, we have $B(x,r) \setminus \{x\} \neq \emptyset$ implying $B(x,r) \nsubseteq \{x\}$ and hence M does not contain any open set. Therefore $\overset{\circ}{M} = \emptyset$ which is closed, thus

$$\overline{\overset{\circ}{M}} = \emptyset \neq M = \overline{M}.$$

(c) <u>Claim:</u> $\overset{\circ}{\overline{M}} \neq \overline{\overset{\circ}{M}}$

We can prove this by considering again the counterexample $M = B(0,1)$. As before $M = \overset{\circ}{M}$, $\overline{\overset{\circ}{M}} = \overline{M} = \{x \in X : \|x\| \leq 1\}$ and $\overset{\circ}{\overline{M}} = \{x \in X : \|x\| < 1\}$. Hence

$$\overset{\circ}{\overline{M}} = \{x \in X : \|x\| < 1\} \neq \{x \in X : \|x\| \leq 1\} = \overline{\overset{\circ}{M}}.$$

Aufgabe 3 (3+5+3 P.)
Untersuchen Sie die folgenden Funktionen auf Stetigkeit:

(a) $f : \mathbb{C} \setminus \{0\} \to \mathbb{C}\,, \qquad f(z) = \dfrac{1}{z},$

(b) $g : \mathbb{R} \to \mathbb{R}\,, \qquad g(x) = \lfloor x \rfloor\, x^2 \quad$ mit $\lfloor x \rfloor := \max\{r \in \mathbb{Z} : r \leq x\},$

(c) (i) $h : \mathbb{R}^2 \to \mathbb{R}\,, \qquad h(x,y) = \begin{cases} \dfrac{x^2+y^2}{|x|+|y|}, & \text{falls} \quad (x,y) \neq (0,0), \\ 0, & \text{falls} \quad (x,y) = (0,0) \end{cases}$

bezüglich der euklidischen Norm $\|(x,y)\| := \sqrt{x^2+y^2}$.

(ii) bezüglichen einer allgemeinen Norm auf $\mathbb{R}^2$.

Lösung. Recall: A function $f : D \to X$, $D \subset X$ is continuous at $x_0 \in D$, if $\forall \varepsilon > 0,\ \exists \delta > 0 :$ $x \in D, \|x - x_0\| < \delta \Rightarrow \|f(x) - f(x_0)\| < \varepsilon$ (Theorem 4.3.2).

(a) Let $0 < \varepsilon < 1$ be arbitrary and $z_0 \in \mathbb{C}\setminus\{0\}$. Assume that $z \in \mathbb{C}$ is such that $|z - z_0| < \delta$. We will decide later how we choose δ. We compute

$$|f(z) - f(z_0)| = \left|\frac{1}{z} - \frac{1}{z_0}\right| = \left|\frac{z_0 - z}{z_0 z}\right| = \frac{|z_0 - z|}{|z_0||z|} < \frac{\delta}{|z_0||z|}. \tag{1}$$

Moreover, we can estimate

$$|z| = |z - z_0 + z_0| \geq ||z - z_0| - |z_0|| > |\delta - |z_0||.$$

Now we see that if we choose $\delta \leq \frac{|z_0|}{2}$, then $|z| > \frac{|z_0|}{2}$. We can use this estimate in (1). Indeed, for $\delta := \varepsilon \min\left\{\frac{|z_0|}{2}, \frac{|z_0|^2}{2}\right\}$ and z with $|z - z_0| < \delta$ we have

$$|f(z) - f(z_0)| < \frac{\delta}{|z_0||z|} \leq \frac{2\delta}{|z_0|^2} \leq \varepsilon.$$

This shows that f is continuous at z_0. Since z_0 was chosen arbitrarily in $\mathbb{C} \setminus \{0\}$, we conclude that f is continuous in $\mathbb{C} \setminus \{0\}$.

(b) In a first step, we show that g is **not** continuous in $\mathbb{Z} \setminus \{0\}$. So let $z \in \mathbb{Z} \setminus \{0\}$. We need to show: $\exists \varepsilon > 0$ such that $\forall \delta > 0\ \exists z_\delta$ with $|z - z_\delta| < \delta$ and $|g(z) - g(z_\delta)| > \varepsilon$. We choose $\varepsilon = \frac{1}{2}$ and let $\delta > 0$ be arbitrary. Moreover, let $n \in \mathbb{N}$, such that $\frac{1}{n} < \delta$ and $n > 4|z-1|(2|z|+1)$ and define $z_\delta = z - \frac{1}{n}$. Clearly, $|z - z_\delta| = \frac{1}{n} < \delta$. Now we compute,

$$\begin{aligned} \left|g(z) - g\left(z - \frac{1}{n}\right)\right| &= \left|z^3 - (z-1)\left(z - \frac{1}{n}\right)^2\right| \\ &= \left|z^3 - (z-1)\left(z^2 - \frac{2z}{n} + \frac{1}{n^2}\right)\right| \\ &= \left|z^2 + \frac{z-1}{n}\left(2z - \frac{1}{n}\right)\right|. \end{aligned}$$

Notice that as $n > 4|z-1|(2|z|+1)$, we have also

$$\left|\frac{z-1}{n}\left(2z - \frac{1}{n}\right)\right| \leq \frac{|z-1|}{n}(2|z| + 1/n) \leq \frac{|z-1|}{n}(2|z|+1) < \frac{1}{4}$$

Thus $\left|g(z) - g\left(z - \frac{1}{n}\right)\right| \geq |z|^2 - \frac{1}{4} \geq \frac{1}{2} = \varepsilon$. With this we proved: g is not continuous in $\mathbb{Z} \setminus \{0\}$.

In the next step, we show: g is continuous in 0. So let $0 < \varepsilon < 1$. For $|z| < \sqrt{\varepsilon} < 1$, we compute

$$|g(0) - g(x)| = |g(x)| = |\underbrace{\lfloor z \rfloor}_{\in\{0,-1\}} z^2| \leq |z|^2 < \varepsilon.$$

This proves that g is continuous in 0.

As a last step, we prove that g is also continuous in $\mathbb{R} \setminus \mathbb{Z}$. So, let $x_0 \in \mathbb{R} \setminus \mathbb{Z}$ and define $m = \lfloor x_0 \rfloor \in \mathbb{Z}$, i.e. $m < x_0 < m+1$. Notice that $\operatorname{dist}(x_0, \mathbb{Z}) = \min\{x_0 - m, (m+1) - x_0\}$. We choose $\delta < \left\{\frac{\varepsilon}{m(2|x_0|+1)}, \frac{\operatorname{dist}(x_0,\mathbb{Z})}{2}\right\}$. Since $\frac{\operatorname{dist}(x_0,\mathbb{Z})}{2} < 1$, we have

$$|x - x_0| < \delta \quad \Longrightarrow \quad (\lfloor x \rfloor = m = \lfloor x_0 \rfloor \text{ and } |x| < |x_0| + 1)\,.$$

With this, we have that for all x with $|x - x_0| < \delta$ the following holds:

$$\begin{aligned} |g(x) - g(x_0)| &= m\left|(x^2 - x_0^2)\right| = m|x - x_0||x + x_0| \\ &< m\delta(2|x_0| + 1) < \varepsilon. \end{aligned}$$

Since $x_0 \in \mathbb{R} \setminus \mathbb{Z}$ was chosen arbitrarily, we conclude: g is continuous in $\mathbb{R} \setminus \mathbb{Z}$.

Alternatively, the continuity of g in $\mathbb{R}\setminus\mathbb{Z}$ can also be shown using Theorem 4.3.5. Indeed, for $x_0 \in \mathbb{R} \setminus \mathbb{Z}$ and $m := \lfloor x_0 \rfloor \in \mathbb{Z}$ we have: $\lfloor x \rfloor = m$ (constant) for all $x \in (m, m+1)$. Therefore, $\hat{g}(x) := \lfloor x \rfloor$ is continuous in $\mathbb{R} \setminus \mathbb{Z}$ (Example 4.3.4 (3) from lecture notes). And, we also know that $\tilde{g}(x) := x^2$ is continuous (in $\mathbb{R}$). Theorem 4.3.5. implies that $g(x) = \hat{g}(x) \cdot \tilde{g}(x)$ is continuous in $\in \mathbb{R} \setminus \mathbb{Z}$.

(c) (i) h is continuous $\in \mathbb{R}^2 \setminus \{(0,0)\}$ as a result of Theorem 4.3.5 (multiplication and addition of continuous functions).

We treat the case $(x_0, y_0) = (0,0)$ separately. Let $\varepsilon > 0$. We choose $\delta = \frac{\varepsilon}{2}$. Then for all $(x,y) \in \mathbb{R}^2$ with $\|(x,y)\| < \delta$ we have

$$\left|\frac{x^2 + y^2}{|x| + |y|} - 0\right| = \frac{x^2}{|x| + |y|} + \frac{y^2}{|x| + |y|} \leq \frac{x^2}{|x|} + \frac{y^2}{|y|} = |x| + |y| \leq 2\|(x,y)\| < \varepsilon.$$

This means that h is also continuous in $(0,0)$.

(ii) Since all norms in $\mathbb{R}$ are equivalent, h is continuous in $\mathbb{R}^2$ with respect to any norm (compare Beispiel 4.3.4 (7)). $\square$

Aufgabe 4 (5+5 P.)

(a) Kann man die Funktion $f : \mathbb{C} \setminus \{0\} \to \mathbb{C}\,,\ f(z) = \dfrac{z}{|z|}$ im Punkt 0 stetig fortsetzen?

(b) Bestimmen Sie den Wert von $\alpha, \beta, \gamma, \delta \in \mathbb{R}$ so, dass die Funktion

$$g(x) = \begin{cases} x^2 - \alpha x + \beta, & \text{falls } x < -1 \\ \gamma, & \text{falls } x = -1 \\ (\alpha + \beta)\, x, & \text{falls } -1 < x < 1\,. \\ \delta, & \text{falls } x = 1 \\ x^2 + \alpha x - \beta, & \text{falls } x > 1 \end{cases}$$

auf der ganzen reellen Achse stetig ist.

Lösung.

(a) Assume (by contradiction) that f is continuously extendable in $z = 0$ and let $\tilde{f}$ be the continuously extended function. Since $\tilde{f}$ is continuous at $z = 0$, the following holds:

$$\forall (z_n)_{n\in\mathbb{N}} \subset \mathbb{C} \setminus \{(0,0)\} \quad \text{with} \lim_{n\to\infty} z_n = 0 \quad \text{it holds} \lim_{n\to\infty} \tilde{f}(z_n) = \tilde{f}(0).$$

In particular, this holds for the two sequences $x_n := \frac{1}{n}$ and $y_n := -\frac{1}{n}$. Then,

$$\text{as} \lim_{n\to\infty} x_n = 0 \quad \text{it holds} \quad \tilde{f}(0) = \lim_{n\to\infty} \tilde{f}(x_n) = \lim_{n\to\infty} \frac{1/n}{1/n} = 1,$$
$$\text{as} \lim_{n\to\infty} y_n = 0 \quad \text{it holds} \quad \tilde{f}(0) = \lim_{n\to\infty} \tilde{f}(y_n) = \lim_{n\to\infty} \frac{-1/n}{1/n} = -1.$$

As $1 \neq -1$, there is no such extension $\tilde{f}$, thus f is **not** continuously extendable.

(b) It is given that the function g is continuous on $\mathbb{R}$. Then according to Theorem 4.3.2, we have $\forall (x_n)_{n\in\mathbb{N}}$ with $\lim_{n\to\infty} x_n = x_0$ it holds $\lim_{n\to\infty} g(x_n) = g(x_0)$. Moreover, g is continuous at $x_0 \in \mathbb{R}$ if and only if it is right and left continuous at x_0 (Theorem 4.3.17). Then,

$$\lim_{x\to x_0^-} g(x) = g(x_0) = \lim_{x\to x_0^+} g(x), \tag{2}$$

where $\lim_{x\to x_0^-} g(x)$ and $\lim_{x\to x_0^-} g(x)$ are the right and left limits.

We compute at $x = -1$

$$\lim_{x\to -1^-} g(x) = \lim_{x\to -1^-} (x^2 - \alpha x + \beta) = 1 + \alpha + \beta$$

and

$$\lim_{x\to -1^+} g(x) = \lim_{x\to -1^+} (\alpha + \beta)x = -\alpha - \beta.$$

As $\lim_{x\to -1^-} g(x) = \lim_{x\to -1^+} g(x)$, it follows $1 + 2\alpha + 2\beta = 0$.

Similarly at $x = 1$ we get,

$$\lim_{x\to 1^-} g(x) = \lim_{x\to 1^-} (\alpha + \beta)x = \alpha + \beta$$

and

$$\lim_{x\to 1^+} g(x) = \lim_{x\to 1^+} (x^2 + \alpha x - \beta) = 1 + \alpha - \beta.$$

Using $\lim_{x\to -1^-} g(x) = \lim_{x\to -1^+} g(x)$, we deduce $\alpha + \beta = 1 + \alpha - \beta$, i.e. $\beta = 1/2$. We substitute for β in $1 + 2\alpha + 2\beta = 0$ and get $\alpha = -1$.

We are left with determining γ and δ. Using equation (1) we can find the value of g at $x = -1$ and $x = 1$:

$$\gamma = g(-1) = \lim_{x\to -1} g(x) = 1 + \alpha + \beta = 1/2,$$
$$\delta = g(1) = \lim_{x\to 1} f(x) = \alpha + \beta = -1/2.$$

□